

# INTRODUCTION TO LPPs AND THE GRAPHICAL METHOD

## 2.1. Introduction

A programming problem is a problem where we have an objective which is to be fulfilled under certain restrictions. The moment this objective and these restrictions are converted into a mathematical expression and mathematical equations or inequalities, respectively, the problem is called a mathematical programming problem. When this mathematical expression and these equations or inequalities are linear in form, that is, the relationship between the variables is linear, in which each variable will only have a power of one and product of variables, logarithmic, exponential, trigonometric, etc. terms are not allowed, then the mathematical programming problem is called a linear programming problem (LPP). Hence, a linear programming problem is a special type of mathematical programming problem which either maximizes or minimizes a linear objective function while satisfying a set of linear equality and/or inequality constraints (restrictions). Mathematical programming problems are optimization problems. In general, a problem where we maximize or minimize a numerical function of variables under some constraints or restrictions which are also in the form of the numerical functions is called an optimization problem. These types of problems may have a number of feasible solutions which satisfy all the constraints, out of which one is selected, which optimizes our objective function.

Any general linear programming problem with  $n$  variables and  $m$  linear constraints can be written as:

$$\text{Maximize or Minimize } Z = c_1x_1 + c_2x_2 + \dots + c_nx_n \text{ (Linear objective function)}$$

Subject to the constraints

$$\begin{aligned}
 &a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n (\leq, =, \geq) b_1 \\
 &a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n (\leq, =, \geq) b_2 \\
 &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 &a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n (\leq, =, \geq) b_i \\
 &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 &a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n (\leq, =, \geq) b_m
 \end{aligned} \tag{2.1}$$

$$x_1, x_2, \dots, x_n \geq 0 \text{ (Non-negative restriction)}$$

The linear inequalities and/or equations represent the restrictions on available resources like man-hours, machine-hours, land, and so on. Here,  $a_{ij}$ ,  $b_i$ ,  $c_j$  where  $i = 1, 2, \dots, m$ ,  $j = 1, 2, \dots, n$  are assumed to be constants. The coefficients  $c_1, c_2, \dots, c_n$  are the unit profits/ cost coefficients of the decision variables  $x_1, x_2, \dots, x_n$ , respectively, in the objective function. The inequality  $\sum_{j=1}^n a_{ij}x_j \leq b_i$  ( $i = 1, 2, \dots, m$ ) represents the  $i^{\text{th}}$  functional constraint (restriction) on the consumption of the  $i^{\text{th}}$  resource where  $a_{ij}$  the coefficient of  $x_j$  ( $j = 1, 2, \dots, n$ ) gives the amount of resource  $i$  consumed by one unit of activity  $j$ ,  $x_j$  is the level of activity  $j$ , and  $b_i$  is the maximum availability of the  $i^{\text{th}}$  resource.

The objective is to determine the non-negative values of the variables  $x_1, x_2, \dots, x_n$  which will satisfy all the constraints on available resources and optimizes the objective function.

(2.1) can also be written as:

$$\begin{aligned}
 &\text{Maximize or Minimize } Z = \sum_{j=1}^n c_j x_j \\
 &\text{Subject to } \sum_{j=1}^n a_{ij}x_j (\leq, =, \geq) b_i, \quad i = 1, 2, \dots, m \\
 &\quad \quad \quad x_j \geq 0, j = 1, 2, \dots, n
 \end{aligned} \tag{2.2}$$

Or in matrix form, the same can be written as:

$$\begin{aligned}
 &\text{Maximize or Minimize } Z = \mathbf{CX} \\
 &\text{Subject to } \mathbf{AX} (\leq, =, \geq) \mathbf{b} \\
 &\quad \quad \quad \mathbf{X} \geq 0
 \end{aligned} \tag{2.3}$$

Here,  $\mathbf{b}_{m \times 1} = (b_1, b_2, \dots, b_m)$ ,  $\mathbf{C}_{(1 \times n)} = (c_1, c_2, \dots, c_n)$ ,  $\mathbf{X}'_{(1 \times n)} = (x_1, x_2, \dots, x_n)$

$$A_{(m \times n)} = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix}$$

**Solution:** An  $n$ -tuple  $(x_1, x_2, \dots, x_n)$  of real numbers which satisfies the constraints of (2.1) is called a solution to the general linear programming problem.

**Feasible Solution:** A solution to a linear programming problem is known as a feasible solution if it also satisfies the non-negative restriction of the problem.

**Basic Feasible Solution:** If a basic solution to the given linear programming problem satisfies the non-negativity conditions also then the solution is called a basic feasible solution. Basic feasible solutions are of two types:

- (i) Degenerate: A basic feasible solution is called degenerate if the value of at least one basic variable is zero.
- (ii) Non-degenerate: A basic feasible solution is called “non-degenerate” if values of all the variables are greater than zero.

**Optimal Basic Feasible solution:** If a basic feasible solution to a given linear programming problem optimizes the objective function, then it is called an optimal basic feasible solution.

*Note.* If a problem has a feasible solution, then it will also have a basic feasible solution.

**Example:** Minimize  $z = x_1 + 2x_2$

Subject to

$$x_1 + 2x_2 \geq 10$$

$$10x_1 + 3x_2 \geq 20$$

$$x_1, x_2 \geq 0$$

This is a minimization problem with two decision variables and two constraints. Here,  $c_1 = 1$ ,  $c_2 = 2$ ,  $b_1 = 10$ ,  $b_2 = 20$ ,  $a_{11} = 1$ ,  $a_{12} = 2$ ,  $a_{21} = 10$ , and  $a_{22} = 3$ .

$x_1 = -1$ ,  $x_2 = 15$  is a solution to the given problem, but it is not a feasible one, as  $x_1 < 0$ .

$x_1 = 0.588235$ ,  $x_2 = 4.705882$  is an optimal basic feasible solution to the given problem, and the corresponding minimum value of  $z$  is 10.

## 2.2. Assumptions in a Linear Programming Problem

To solve any real-life problem by the linear programming approach (LP), the following assumptions are made:

**1. Certainty.** This assumption requires that all the parameters of the model be known before solving a given LP problem. In other words, the coefficients  $c_j$  of the objective function, the coefficients of the constraints  $a_{ij}$ , and the resource vector  $b_i$  must all be known and fixed. However, using sensitivity analysis (which will be discussed in Chapter 6), the decision maker can explore the effects of changing some parameters over a range of values.

**2. Linearity.** This is a property of the mathematical function. Linearity denotes a stable relationship between a dependent and independent variables, which when plotted on a graph would be represented by a straight line. In linear programming, the relationship of the various decision variables either in the objective function or in any of the constraints should be linear. Note that in a linear relationship, all variables have an exponent of one, and no variables are put in product form.

The following are three properties which help us in clarifying the implications of linearity:

- (i) **Proportionality.** This requires that the value of the objective function be proportional to the value of the decision variables. For example, in an objective function  $Max\ 25x_1 + 38x_2$ , each unit of  $x_1$  should contribute 25 (a proportionality constant) to the value of the objective function, and hence the total contribution of any activity in the objective function depends on the level of activity or the value of the decision variable. In the constraints, the amount of each resource used must be proportional to the value of the decision variable. For example, let  $x_1 + 4x_2 \leq 50$  be any constraint of a given LPP, and then each value of  $x_2$  will consume 4 units (a proportional amount) of the available resource (in this case 50).
- (ii) **Additivity.** Linearity also requires that the effect of the value of each variable on the values of the objective function and the constraints is additive. In other words, there can be no interactions between the effects of different activities. For some objective function, for example,  $Max\ 25x_1 + 38x_2$ , if  $x_1 = 2$  and  $x_2 = 1$ , this property asserts that the contributions of 50 and 38, respectively, must add together to give a sum of 88.

(iii) **Divisibility (or continuity).** This postulate allows the variables to have a continuous range of values. This means that the variables may take any fractional or decimal value and will not necessarily be a whole number or integer. However, in many real-life situations, there may be cases where the variables are allowed to have only integer values, and in such situations LP may be used to provide an approximate answer by rounding off the solution to the nearest integer value, but doing so may cause a significant departure from optimality. If integer solutions are required, then the given linear programming problem is a special type of problem called a linear integer programming problem (to be discussed later).

### 2.3. Theorems on Extreme Points

**Theorem 2.1.** The set of feasible solutions to a given LPP forms a convex set.

**Proof.** Let  $S$  denote the set of feasible solutions to the following LPP,

$$\text{Max } Z = CX$$

Subject to

$$AX \leq b$$

$$X \geq 0 \quad \dots(2.4)$$

$$\text{where } S = \{X \mid AX \leq b, X \geq 0\}$$

Let  $X_1, X_2 \in S$ , such that

$$AX_1 \leq b \quad \dots(2.5)$$

$$AX_2 \leq b \quad \dots(2.6)$$

Let  $X'$  denote the point on the line segment joining  $X_1$  and  $X_2$ , then

$$X' = \lambda X_1 + (1 - \lambda)X_2, \quad 0 \leq \lambda \leq 1 \quad \dots(2.7)$$

Multiplying both sides of (2.7) by  $A$ , we get

$$AX' = \lambda AX_1 + (1 - \lambda)AX_2 \leq \lambda b + (1 - \lambda)b \quad (\text{From (2.5) and (2.6)})$$

$$\Rightarrow AX' \leq b$$

which implies that  $X' \in S$ . Hence, the set of feasible solutions to a given LPP is a convex set.

**Theorem 2.2.** If the set of feasible solutions to a given LPP is a convex polyhedron, then at least one extreme point of the problem will be the optimum solution of the LPP.

**Proof.** Let,

$$\begin{aligned} \text{Max } z &= \mathbf{c}\mathbf{x}, \\ \text{Subject to } A\mathbf{x} &= \mathbf{b}, \mathbf{x} \geq 0 \end{aligned} \quad \dots(2.8)$$

be a given LPP and  $S$  be a set of feasible solutions to (2.8), which is a convex polyhedron. Also let  $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$  denote the extreme points of  $S$  and  $\mathbf{x}^* \in S$  be the optimum solution of (2.8), that is,  $\mathbf{z}^* = \mathbf{c}\mathbf{x}^* \geq \mathbf{c}\mathbf{x} \forall \mathbf{x}$ .

If  $\mathbf{x}^*$  is not an extreme point of  $S$ , then

$$\mathbf{x}^* = \sum_{i=1}^k \lambda_i \mathbf{x}_i, \lambda_i \geq 0 \text{ and } \sum_{i=1}^k \lambda_i = 1 \quad \dots(2.9)$$

$$\text{This implies } \mathbf{z}^* = \mathbf{c}\mathbf{x}^* = \sum_{i=1}^k \lambda_i \mathbf{c}\mathbf{x}_i \quad \dots(2.10)$$

Let  $\max \mathbf{c}\mathbf{x}_i = \mathbf{c}\mathbf{x}_p \forall i = 1, 2, \dots, k$ . Now (2.10) can be written as

$$\mathbf{c}\mathbf{x}^* \leq \left( \sum_{i=1}^k \lambda_i \right) \mathbf{c}\mathbf{x}_p \Rightarrow \mathbf{c}\mathbf{x}^* \leq \mathbf{c}\mathbf{x}_p \text{ since } \sum_{i=1}^k \lambda_i = 1$$

But  $\mathbf{c}\mathbf{x}^*$  is the maximum value of  $z$ , hence  $\mathbf{c}\mathbf{x}^* = \mathbf{c}\mathbf{x}_p$

$$\Rightarrow \mathbf{x}^* = \mathbf{x}_p \text{ (one of the extreme points).}$$

Hence, the optimum solution of the LPP is attained at one of the extreme points.

**Theorem 2.3.** Let  $S = \{\mathbf{x} \in \mathbf{R}^n : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$  be the set of feasible solutions of the following LPP,

$$\begin{aligned} \text{Max } z &= \mathbf{c}\mathbf{x}, \\ \text{Subject to } \\ A\mathbf{x} &= \mathbf{b}, \mathbf{x} \geq 0 \end{aligned}$$

Then  $\mathbf{x}^* \in S$  is a basic feasible solution of the given problem if and only if it is an extreme point of  $S$ .

**Proof. (Part 1)** Let  $\mathbf{x}^* \in S$  be a basic feasible solution of the given problem, such that

$$A\mathbf{x}^* = \mathbf{b}, \mathbf{x}^* \geq 0 \quad \dots(2.11)$$

We have to prove that this is an extreme point; that is, we have to show that two points do not exist in the set such that  $x^*$  can be written as a convex combination of these points.

On the contrary, let us assume that  $x^*$  is not an extreme point of the set of feasible solutions, and then two solutions  $x'$  and  $x'' \in S$  exist, such that

$$x^* = \lambda x' + (1 - \lambda)x'', \quad 0 < \lambda < 1 \quad \dots(2.12)$$

$$\text{Further, let } x^* = [x_B^*, x_{NB}^*], x' = [x_B', x_{NB}'], x'' = [x_B'', x_{NB}''] \quad \dots(2.13)$$

where  $x_B^*, x_B'$  and  $x_B''$  are  $m$  component vectors and  $x_{NB}^*, x_{NB}'$ , and  $x_{NB}''$  are  $(n - m)$  component vectors.

From (2.12) and (2.13), we have

$$[x_B^*, x_{NB}^*] = \lambda[x_B', x_{NB}'] + (1 - \lambda)[x_B'', x_{NB}''] \quad \dots(2.14)$$

Since  $x^* \in S$  is a basic feasible solution, we have  $x_{NB}^* = 0$  and  $Bx_B^* = b$  where  $B$  is a  $m \times m$  order basic matrix for the given LPP.

Now (2.14) can be written as

$$0 = \lambda x_{NB}' + (1 - \lambda)x_{NB}'' \text{ where } 0 < \lambda < 1 \Rightarrow x_{NB}' = x_{NB}'' = 0 \quad \dots(2.15)$$

$$\text{Further, } Ax' = Bx_B' = b \text{ and } Ax'' = Bx_B'' = b \Rightarrow x_B^* = x_B' = x_B''$$

and hence  $x^* = x' = x''$  which proves that our assumption is wrong and  $x^*$  is an extreme point of  $S$ .

**(Part 2)** Here we will show that if  $x^* \in S$  is an extreme point, then it is a basic feasible solution to the given LPP.

Here,  $x^*$  is an  $n$  component vector. Let us assume that its  $r$  components are positive and the rest are zero such that

$$\sum_{j=1}^r a_j x_j^* = b, \quad x_j^* > 0, \quad j = 1, 2, \dots, r \quad \dots(2.16)$$

In order to prove that  $x^*$  is a basic feasible solution, we have to show that the vectors associated with the positive components of  $x^*$  are linearly independent. To prove it, let us assume on the contrary that the vectors associated with the positive components of  $x^*$  are linearly dependent, that is:

$$\sum_{j=1}^r \beta_j a_j = 0 \text{ in which at least one } \beta_j \text{ is non-zero} \quad \dots(2.17)$$

$$\text{Now consider } \theta = \min \left[ \frac{x_j^*}{|\beta_j|}, \beta_j \neq 0, j = 1, 2, \dots, r \right] \text{ positive number}$$

If we take  $0 < \varepsilon < \theta$ , then  $x_j^* - \varepsilon\beta_j > 0$ , and  $x_j^* - \varepsilon\beta_j > 0, j = 1, 2 \dots r \dots$  (2.18)

In the vector form, (2.18) can be written as

$x' = x^* + \varepsilon\beta$  and  $x'' = x^* - \varepsilon\beta$  where  $x^*, x', x'', \beta$  are  $n$  vector components. ... (2.19)

Also we have  $x', x'' \geq 0$ , that is, both are feasible solutions, and hence we have

$Ax' = Ax^* + \varepsilon A\beta = Ax^* = b$  (since  $A\beta = 0$  from (2.17)). Similarly,  $Ax'' = Ax^* - \varepsilon A\beta = Ax^* = b$

Further from (2.19) we can write  $x^* = \left(\frac{1}{2}\right)x' + \left(\frac{1}{2}\right)x''$ , which contradicts the fact that  $x^*$  is an extreme point. Hence, our assumption is wrong.

It proves that the vectors associated with the positive variables are linearly independent, and hence the extreme point is a basic feasible solution.

**Theorem 2.4.** If an optimum value of the objective function occurs at any two extreme points of the feasible region, then all the points on the line segment joining these points also gives the optimal value of the objective function.

**Proof.** Let the maximum value of the objective function of the following LPP,

$$\text{Max } z = cx,$$

Subject to

$$Ax = b, x \geq 0$$

occur at the extreme points  $x_1$  and  $x_2$ , and then let  $\text{Max } z = z^* = cx_1 = cx_2$ . Let  $x'$  be the point on the line segment joining  $x_1$  and  $x_2$ , and then

$$x' = \lambda x_1 + (1 - \lambda)x_2, 0 \leq \lambda \leq 1$$

$$\Rightarrow cx' = \lambda cx_1 + (1 - \lambda)cx_2 = z^*$$

$$\Rightarrow x' \text{ is also an optimal solution of the problem and hence the theorem.}$$

## 2.4. Areas of Application of LPPs

**I. Product Mix Problem.** Here we have to determine the right mix and quantities of products to be manufactured so as to maximize the profit. A firm can always produce different types of products which require different resources in different quantities, and each product is contributing a certain



amount towards profit. The final product mix must take into consideration the limited resources, expected demand for each product, and various policies of the company, and at the same time maximize the profit.

**II. Blending Problem.** In this type of problem, several raw materials are mixed into the final product that must fulfill certain specifications. Each of the raw materials contributes certain properties to the final product and involves a certain cost.

**III. Media Allocation Problem.** Suppose a company has a certain fixed budget which is to be utilized for the advertisement of a particular product in different media where each medium has a different impact and reach. The cost of insertion of an advertisement into each medium is different and known. The problem is to determine the number of allocations of an advertisement in different media so that there is a maximum exposure of the product to its target populations, which maximizes its sales.

**IV. Production Scheduling and Inventory Planning.** Suppose the demand for a particular product fluctuates over a period of time. The varying production rates have proven to be costly. The problem is to determine a smooth production schedule that meets the anticipated demand and yet maintains a reasonable inventory level that minimizes the overall production and inventory carrying cost.

**V. Purchasing Problem.** Linear programming can also be used in deciding about the purchase of various products which are available at different discounts and prices for different quantity ranges. The problem is to determine the quantity of different products that should be purchased so as to maximize (minimize) the overall profit (cost).

**VI. Assignment and Transportation Problem.** The transportation and assignment problem can always be formulated as a special type of linear integer programming problem, satisfying the various row and column requirements and finally minimizing the cost of assignment or transportation.

## 2.5. Formulation of Linear Programming Models

The following steps need to be taken for a successful formulation of an LP model:

- I. Understand the problem.
- II. Identify the decision variables involved in the problem.

- III. Identify the various coefficients of the objective function.
- IV. Identify the various restrictions of the problem.
- V. Represent the objective function in the form of a linear expression involving decision variables.
- VI. Represent all the restrictions in the form of linear expressions involving decision variables.

**Example 2.1. (Product Mix)** A small plant manufactures two models of a product, say A and B. Two resources, R1 and R2, are required for the manufacturing of units of this product. One unit of A consumes 10 units of R1 and 15 units of R2. Similarly, one unit of B consumes 12 units of R1 and 17 units of R2. Suppose further that the company has 800 units of R1 and 600 units of R2 available per week, and each unit of model A and B gives a profit of \$50 and \$70, respectively. Formulate this as a linear programming problem so as to find the optimal weekly production levels for these two models.

**Solution:** Let  $x_1$  and  $x_2$  denote the weekly production levels of model, A and model B, respectively.

The LP model of the given problem is,

$$\text{Max } z = 50x_1 + 70x_2 \text{ (Objective function)}$$

Subject to

$$10x_1 + 12x_2 \leq 800 \text{ (R1 resource)}$$

$$15x_1 + 17x_2 \leq 600 \text{ (R2 resource)}$$

$$x_1, x_2 \geq 0 \text{ (non-negative condition)}$$

**Example 2.2. (Production Scheduling)** A firm manufactures three different versions of the same product, which have unit profit contributions of 2, 3, and 3.5. In the final part of the manufacturing process, there are assembly, polishing, and packing operations. For each version, the time (in hrs.) required for these operations is shown in the following table:

|                  |   | Assembly | Polishing | Packing |
|------------------|---|----------|-----------|---------|
| Product versions | 1 | 4        | 3         | 3       |
|                  | 2 | 5        | 1         | 4       |
|                  | 3 | 2        | 4         | 1.5     |

*Under the given state of the labor force, the firm estimates that each year, they have 1,000 hours of assembly time, 800 hours of polishing time, and 950 hours of packing time available. Formulate a linear programming problem to find the number of each version the company should make per year in order to maximize the associated profit.*

**Solution:** Let  $x_1$ ,  $x_2$ , and  $x_3$  be the number of units of the first, second, and third version of the product produced per year, respectively. Then the LP model of the given problem is,

$$\text{Max } z = 2x_1 + 3x_2 + 3.5x_3 \text{ (Objective function)}$$

*Subject to*

$$4x_1 + 5x_2 + 2x_3 \leq 1000 \text{ (Assembly)}$$

$$3x_1 + x_2 + 4x_3 \leq 800 \text{ (Polishing)}$$

$$3x_1 + 4x_2 + 1.5x_3 \leq 950 \text{ (Packing)}$$

$$x_1, x_2, x_3 \geq 0 \text{ (Non-negativity condition)}$$

**Example 2.3. (Purchasing Problem)** *A firm is planning to buy two types of cupboards for its new office room, which has a maximum area of 110 square feet for the cupboards. The first type of cupboard costing \$150 per unit requires five square feet of floor space, and holds nine cubic feet of files, whereas the second cupboard costs \$170 per unit, requires seven square feet of floor space, and holds eleven cubic feet of files. The firm has a total budget of \$10,000 for this purpose. How many of which type of cupboard should the firm buy in order to maximize storage volume?*

**Solution:** Let  $x_1$  and  $x_2$  be the number of units of the first and second type of cupboards, respectively. Then the objective function of the problem is to maximize storage volume of the office room subject to the cost and space constraints. So, the LP model of the given problem is,

$$\text{Max } z = 9x_1 + 11x_2$$

*Subject to*

$$150x_1 + 170x_2 \leq 10000 \text{ (Cost Restriction)}$$

$$5x_1 + 7x_2 \leq 110 \text{ (Floor Restriction)}$$

$$x_1, x_2 \geq 0 \text{ (Non-negativity condition)}$$

**Example 2.4. (Transportation Problem)** *A consumer durable company which manufactures a single product has three plants, I, II, and III.*

The amounts of units produced in these plants are 100, 150, and 120, respectively. The firm has its commitment to sell 80 units to customer A, 100 to B, 70 to C, and 120 to D. The net unit cost of transportation to these four different customers from the three plants is given as follows:

|        |     | Customers |    |    |    |
|--------|-----|-----------|----|----|----|
|        |     | A         | B  | C  | D  |
| Plants | I   | 10        | 20 | 15 | 18 |
|        | II  | 21        | 10 | 19 | 11 |
|        | III | 18        | 14 | 22 | 13 |

Formulate the previous transportation problem as a linear programming problem so as to minimize the total cost of transportation.

**Solution:** Let  $x_{ij}$  ( $i = 1, 2, 3$  and  $j = 1, 2, 3, 4$ ) denote the amount of units transported from the  $i^{\text{th}}$  plant to the  $j^{\text{th}}$  customer. Then the objective function of the problem is,

$$\begin{aligned} \text{Min } z = & 10x_{11} + 20x_{12} + 15x_{13} + 18x_{14} + 21x_{21} + 10x_{22} \\ & + 19x_{23} + 11x_{24} + 18x_{31} + 14x_{32} + 22x_{33} + 13x_{34} \end{aligned}$$

Subject to

**Row Constraints:**

$$\sum_{j=1}^4 x_{1j} = 100$$

$$\sum_{j=1}^4 x_{2j} = 150$$

$$\sum_{j=1}^4 x_{3j} = 120$$

**Column Constraints:**

$$\sum_{i=1}^3 x_{i1} = 80$$

$$\sum_{i=1}^3 x_{i2} = 100$$

$$\sum_{i=1}^3 x_{i3} = 70$$

$$\sum_{i=1}^3 x_{i4} = 120$$

and  $x_{ij} \geq 0, \forall i, j$  (**non-negativity condition**)

**Example 2.5. (Media Allocation Problem)** A company has an advertising budget of \$10,000,000 and is planning to advertise its products in three different media, namely television, radio, and newspapers. The advertising in television is further divided into prime time and non-prime time. The objective of the company is to have maximum exposure of the product to its potential customers. A survey of the market gives the following results:

|                                                | Television<br>Prime<br>time | Non-prime<br>time | Radio   | News<br>paper |
|------------------------------------------------|-----------------------------|-------------------|---------|---------------|
| Cost of an Advt./unit(\$)                      | 50,000                      | 28,000            | 20,000  | 19,000        |
| No. of potential customers reached/unit        | 700,000                     | 350,000           | 250,000 | 180,000       |
| No. of female potential customers reached/unit | 500,000                     | 100,000           | 110,000 | 87,000        |

It was also decided that the company will spend no more than 50% of the total budget in television. The management was also not willing to have more than 7 and 10 insertions in radio and newspapers respectively. Finally, the company has also decided to have a minimum of 1.5 million exposures among women. Formulate this problem as an LP model so as to maximize the total reach of the potential customer.

**Solution:** Let  $x_1$ ,  $x_2$ ,  $x_3$ , and  $x_4$  denote the number of insertions in prime time, non-prime time (TV), radio, and newspapers respectively. The objective of the company is to maximize the total reach of the potential customer. Then the LP model is:

$$\text{Max } z = 700000x_1 + 350000x_2 + 250000x_3 + 180000x_4$$

**(Maximization of potential reach)**

*Subject to*

$$50000x_1 + 28000x_2 + 20000x_3 + 19000x_4 \leq 1,000,000,$$

**(Budget constraint)**

$$50000x_1 + 28000x_2 \leq 500,000,$$

**(Restriction on Television advertisement)**

$$500000x_1 + 100000x_2 + 110000x_3 + 87000x_4 \geq 1,500,000$$

**(Restriction on the minimum number female potential customers to be reached by the advertisement)**

$$x_3 \leq 7, x_4 \leq 10$$

(Upper limits on the number of insertions in Radio and Newspapers)

$$x_1, x_2, x_3, \text{ and } x_4 \geq 0$$

## 2.6. Graphical Method

The graphical method can be used to solve any LPP with two variables only. There are two different approaches for solving an LPP using the graphical method:

### 2.6.1. Extreme Point Approach

The following steps need to be followed while using this approach:

- I. First, draw an  $XY$  plane and select the set of points  $(x, y)$  which satisfy the non-negative restrictions of the given LPP. This set of points will be a set of feasible solutions to the given problem. All these points in the set of feasible solutions lie in the first quadrant of the  $xy$  plane.
- II. Draw all the constraints in the  $xy$  plane and identify the common region, that is, a set of solutions  $(x, y)$  which satisfies all the constraints and lie in the first quadrant. This set of solutions is called a common region of feasible solutions.
- III. Identify the corner points (extreme points) of the common region of feasible solutions; find the value of the objective function at these points. Select the point which optimizes the objective function, and this will represent our optimal solution.
- IV. Give your interpretation to the result.

**Note:** The common feasible region of any LP problem always forms a convex set, and its optimum solution lies at one of the extreme points of the convex set.

### 2.6.2. ISO-Profit (cost) Function Line Approach

After doing the first two steps (I & II) of the procedure given in 2.6.1, continue with the following:

- I. Draw an ISO-profit (ISO-cost) line for a small value of the objective function without violating any of the given restrictions of the given LPP.

- II. Continue drawing ISO-profit (ISO-cost) lines parallel to one another in the direction of increasing (or decreasing) the value of the objective function.
- III. The last point of the feasible region where this line will touch before it moves out of the (feasible) region for which the value of ISO-profit (ISO-cost) is maximum (minimum) is the optimal solution.

We will discuss the steps involved in solving a simple linear programming model graphically with the help of an example.

In this book, our main focus will be on the extreme point approach.

## 2.7. Solution of LPPs by the Graphical Method Using MS-Excel

**Step 1.** Select the xy-values which are to be plotted and then go to **Insert** → **Scatter** → **Scatter with straight line and Markers**.

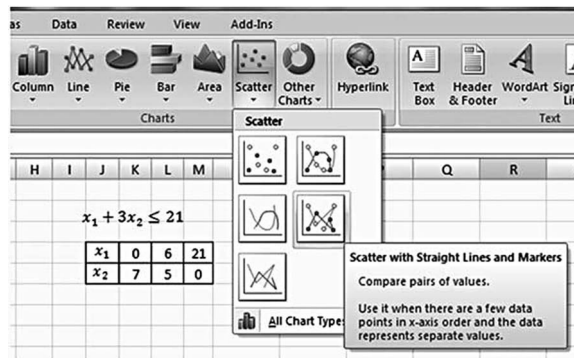


Fig. 2.1

**Step 2.** Right-click on the chart and click the **Select Data** option.

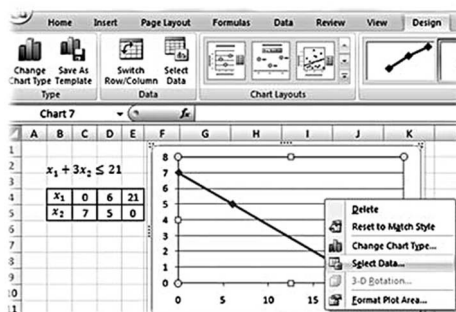


Fig. 2.2

**Step 3.** In the **Legend Entries (series)**, select the **Add** option and add all the constraints of the problem.

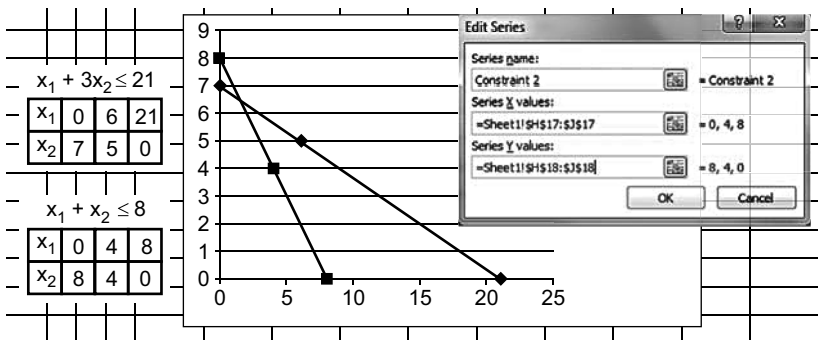
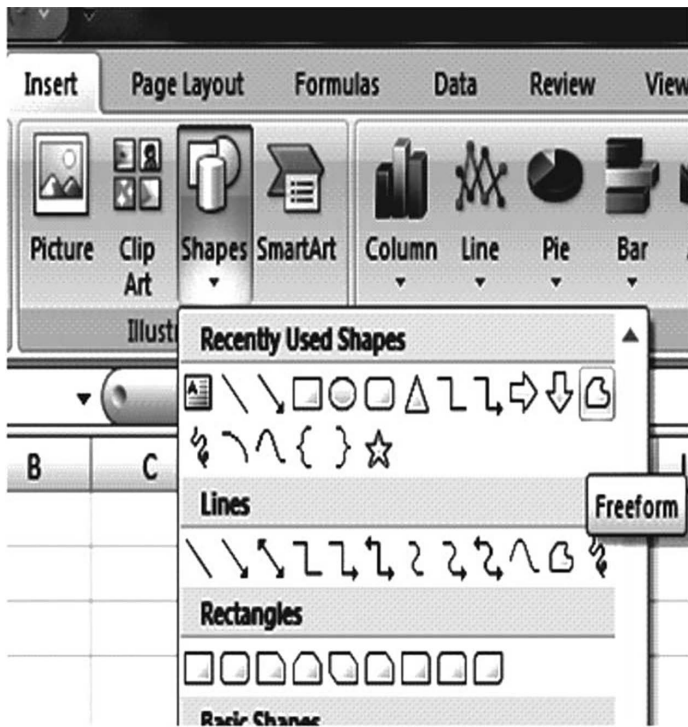


Fig. 2.3

**Step 4.** Identify the common portion to all the constraints. For shading this region go to **Insert** → **Shapes** → **Freeform** and then click the cursor to the corner points of the feasible region.





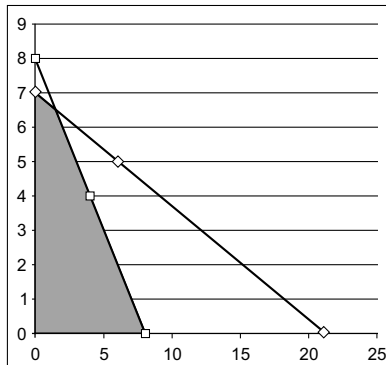


Fig. 2.4

**Step 5.** For the formatting of the chart, such as inserting text, gridlines, legends, and so on, double-click on the chart and select the **Layout** option.

**Example 2.6.** Maximize  $z = 2x_1 + 4x_2$

Subject to the constraints:

$$x_1 + 3x_2 \leq 21$$

$$x_1 + x_2 \leq 8$$

$$x_1 \leq 3$$

$$x_1, x_2 \geq 0$$

**Solution:** In the  $x_1x_2$ -plane, we take  $x_1$  on the  $x$ -axis and  $x_2$  on the  $y$ -axis. When the two constraints along with the non-negativity conditions are plotted in the  $x_1x_2$  plane, we have the following figure.

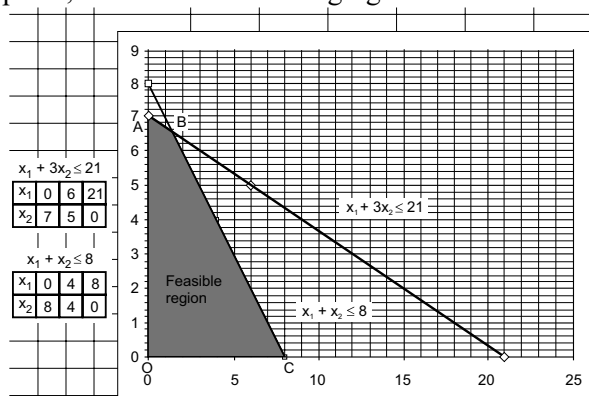


Fig. 2.5

**Extreme point approach:** In the previous graph the shaded region denoted by **OABC** is the feasible region for the given problem. We also know that the optimum solution of the problem lies at one of the extreme points of this region. So, the value of the objective function  $z$  at these extreme points is:

- I. at the point O (0, 0),  $z = 0$
- II. at the point A (0, 7),  $z = 28$
- III. at the point B (1.5, 6.5),  $z = 29$
- IV. at the point C (8, 0),  $z = 16$

We can clearly see that  $z = 29$  is the maximum value of  $z$ , and this occurs at point B. Hence, the optimum solution of the problem is  $(x_1, x_2) = (1.5, 6.5)$ , and the corresponding maximum value of the objective function is  $z^* = 29$ .

**ISO-profit (cost) function line approach:** To get an **iso profit line**, set the objective function equal to some arbitrary small number. This will give us a linear equation in  $x$  and  $y$ , and that can be plotted in the  $xy$ -plane. All the points on an **iso profit line** give the same profit. To start with, let us put  $z = 10$ . Then the equation becomes  $2x_1 + 4x_2 = 10$ , and this line contains all the points that have a profit of exactly 10.

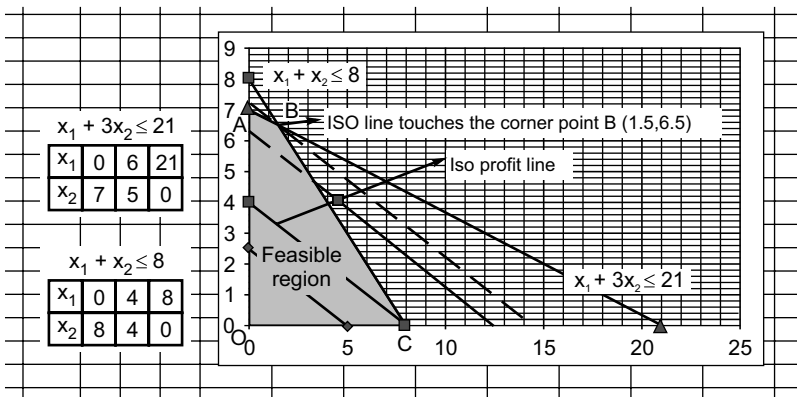


Fig. 2.6

Now find the highest value of the **iso profit line** that touches the feasible region, which is obtained by plotting a number of parallel lines, away from the origin until it touches the corner point(s) of the feasible region. Here the **iso profit line** touches the corner point **B** for the first time. The coordinates of

corner points can be read from the graph or can be computed as the intersection of the two linear equations.

Therefore, the optimum solution of the problem is  $x_1 = 1.5$ ,  $x_2 = 6.5$ , and the corresponding maximum value of the objective function is  $z^* = 29$ .

**Example 2.7.** Minimize  $z = 25x_1 + 38x_2$

Subject to the constraints:

$$30x_1 + 10x_2 \geq 120$$

$$10x_1 + 20x_2 \geq 100$$

$$x_1, x_2 \geq 0$$

**Solution:** When the two constraints along with the non-negativity conditions are plotted in the  $x_1$   $x_2$  plane, we have the following figure.

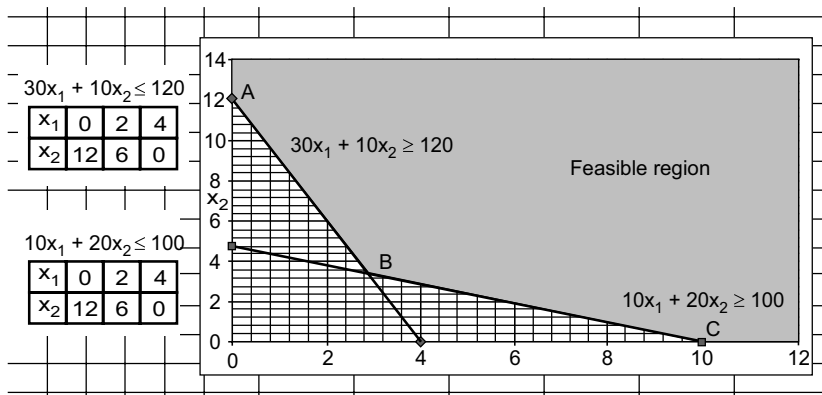


Fig. 2.7

The area inside the region ABC is the feasible region of the problem, and the points A, B, and C denote the extreme points of the region. The coordinates of the extreme points are: A = (0, 12), B = (2.8, 3.6), and C = (10, 0).

The values of the objective function at these points are as follows:

- I. at the point A (0, 12),  $z = 456$
- II. at the point B (2.8, 3.6),  $z = 206.8$
- III. at the point C (10, 0),  $z = 250$

Hence, the optimum solution of the problem is  $(x_1 = 2.8, x_2 = 3.6)$ , and the corresponding minimum value of the objective function is  $z^* = 206.8$ .

**Example 2.8.** Maximize  $z = 4x_1 + x_2$

Subject to the constraints:

$$2x_1 + x_2 \leq 100$$

$$x_1 + x_2 \leq 80$$

$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

**Solution:** When the three constraints along with the non-negativity conditions are plotted in the  $(x_1, x_2)$  plane, we have the following figure:

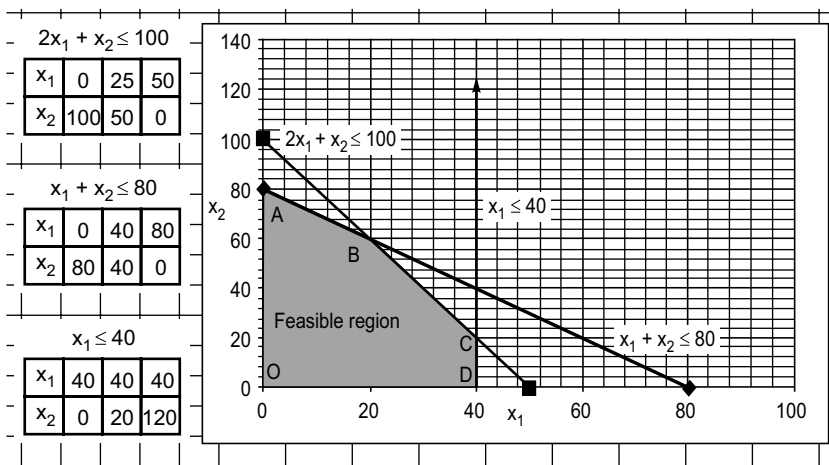


Fig. 2.8

The shaded region OABCD in the previous figure gives the feasible region of the problem, and the values of  $z$  at these points (extreme) are as follows:

- I. at the point O (0, 0),  $z = 0$
- II. at the point A (0, 80),  $z = 80$
- III. at the point B (20, 60),  $z = 140$
- IV. at the point C (40, 20),  $z = 180$
- V. at the point D (40, 0),  $z = 160$

Clearly  $z = 180$  is the maximum value  $z$ , and this occurs at the point C (40, 20). Hence, the optimum solution of the problem is  $(x_1 = 40, x_2 = 20)$ , and the corresponding maximum value of the objective function is  $z^* = 180$ .

The following section discusses some special cases in linear programming problems.

## 2.8. Special Cases

### 2.8.1. Problem with Multiple Solutions

In some of the linear programming problems, instead of having a unique optimal basic feasible solution to the given problem, we may have an alternative or multiple (infinite) numbers of optimal basic feasible solutions. This case occurs when the line of the objective function is parallel to any one of the constraint lines and lies on one of the edges of the region of feasible solutions, which is illustrated in the following example:

**Example 2.9.** Maximize  $z = x_1 + 2x_2$

Subject to the constraints:

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2, x_1, x_2 \geq 0$$

**Solution:** When the constraints along with the non-negativity conditions are plotted in the  $x_1$   $x_2$  plane, we have the following figure:

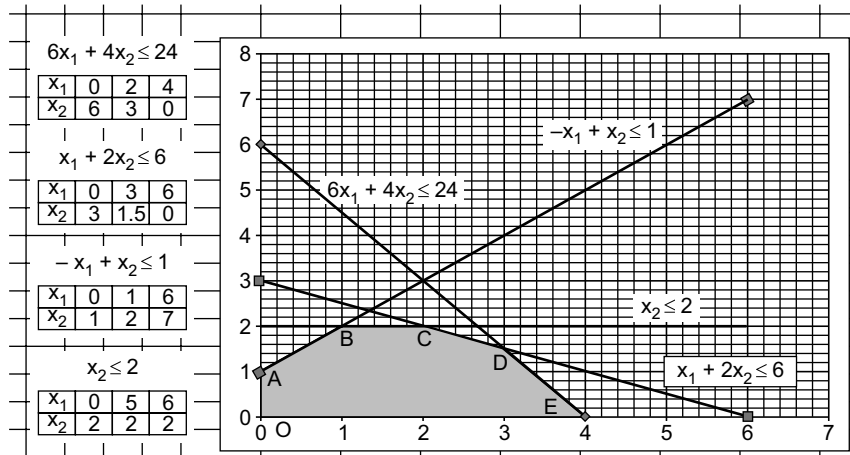


Fig. 2.9

The area inside OABCDE denotes the feasible region of the problem. The values of  $z$  at these points are:

- I. at the point O (0, 0),  $z = 0$
- II. at the point A (0, 1),  $z = 2$
- III. at the point B (1, 2),  $z = 5$
- IV. at the point C (2, 2),  $z = 6$
- V. at the point D (3, 1.5),  $z = 6$

The maximum value of  $z$  is 6, and this occurs at two points, C and D. Therefore, it's a case of multiple solutions. All the points which lie on the line segment joining these two points will also give the same value of  $z$ .

***Note:** If a problem has two points giving the same optimum value of the objective function, then this problem will have an infinite number of solutions (all the points on the line segment joining these two points), and all solutions will give the same value of the objective function.*

### 2.8.2. The Problem with Unbounded Solutions

In some problems, the solution space may be unbounded in at least one variable; that is, the value of that variable may be increased indefinitely without violating any of the given constraints, thus giving an infinitely large or small value of the objective function. This kind of problem has an unbounded solution. The occurrence of unbounded solution space is the indication that the problem is poorly designed. The following example will be used to understand it better.

**Example 2.10.** Maximize  $z = 4x_1 + 10x_2$

Subject to the constraints:

$$\begin{aligned}x_1 &\leq 10 \\x_1 + 2x_2 &\geq 6 \\x_1, x_2 &\geq 0\end{aligned}$$

**Solution:** The solution space satisfying the constraints and the non-negativity restrictions is shown shaded in the following figure. It can be seen that in the solution space the value of  $x_2$  can be increased infinitely because of the second constraint,  $x_1 + 2x_2 \geq 6$ ; due to this the value of objective function also can be increased infinitely.

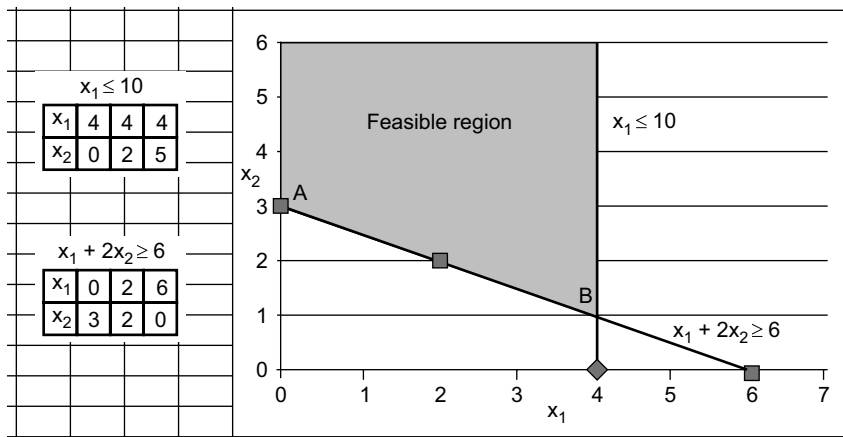


Fig. 2.10

Hence, the value of  $z$  can be made arbitrarily large by increasing the value of  $x_2$  indefinitely, which implies that the given LP problem has an unbounded solution.

### 2.8.3. The Problem with Inconsistent Constraints

The constraints of a given problem are said to be inconsistent if they are mutually exclusive, that is, if no point or solution satisfies all of the constraint equations simultaneously and hence there is no unique common region of feasible solutions. Sometimes a common region of solutions exists but none of the points of this region satisfies the non-negative restrictions of the given LPP. In both these situations, there is no solution for the given problem. Such a problem arises due to wrong or improper formulation of the problem with conflicting constraints.

The problem with inconsistent constraints is said to be an infeasible problem. Infeasibility is purely due to the constraints of the problem the objective function has no role in it.

**Example 2.11.** Maximize  $z = x_1 + 4x_2$

Subject to the constraints:

$$x_1 + 2x_2 \geq 40$$

$$x_1 + x_2 \leq 15$$

$$x_1, x_2 \geq 0$$

**Solution:** When the two constraints along with the non-negativity conditions are plotted in the  $x_1x_2$  plane, we have the following figure:

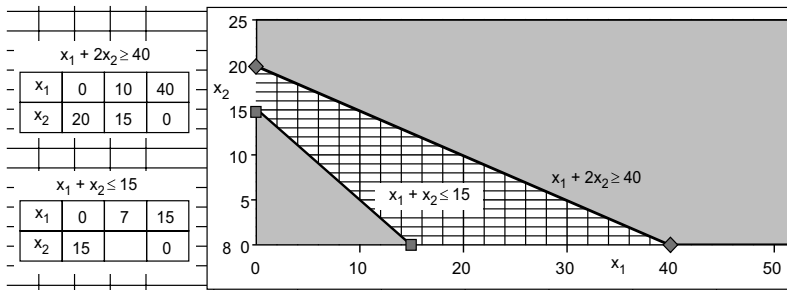


Fig. 2.11

It can be clearly seen from the figure that there is no common region between the two constraints and hence no feasible solution for the problem. Such a problem is said to be a problem with inconsistent constraints, and the problem does not possess any solution.

#### 2.8.4 The Problem with Redundant Constraint Equations

Any LPP is said to have a redundant constraint if at least one of the constraints does not play any role in finding the optimal solution of the given problem, and if even when it is (or they are) removed from the problem, we get the same solution as was obtained when it was (or they were) present in the problem.

**Example 2.12.** Maximize  $z = 2x_1 - x_2$

Subject to the constraints:

$$x_1 + 4x_2 \leq 50$$

$$x_1 + x_2 \leq 20$$

$$1.5x_1 + x_2 \leq 30 \text{ and } x_1, x_2 \geq 0$$

**Solution:**

The area inside OABC denotes the feasible region of the problem. The values of  $z$  at these points are:

- I. at the point O (0, 0),  $z = 0$
- II. at the point A (0, 12.5),  $z = -12.5$
- III. at the point B (10, 10),  $z = 10$
- IV. at the point C (20, 0),  $z = 30$



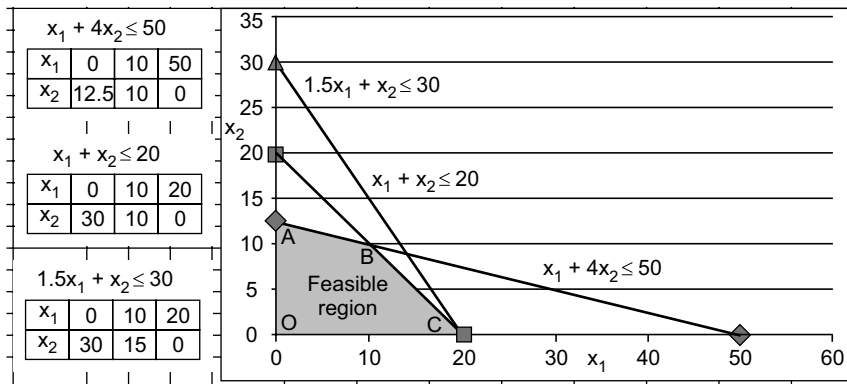


Fig. 2.12

The maximum value of  $z$  is 30, and this occurs at the point C. It can be clearly seen that with or without the third constraint, the optimal solution of the problem will remain the same. Hence, the constraint  $1.5x_1 + x_2 \leq 30$  is said to be redundant.

## Exercises

1. What is a linear programming problem? What are its limitations? Discuss briefly its various areas of application.
2. Write at least five applications of linear programming problems.
3. Explain why solutions to linear programming problems focus on corner points.
4. In the context of a linear programming problem, define the following;
  - (i) Solution
  - (ii) Feasible solution
  - (iii) Basic feasible solution and extreme points of a feasible region
  - (iv) Optimal basic feasible solution
5. What is a feasible region of a linear programming problem? When is it solved using the graphical method?
6. Give the generalized definition of an LPP.
7. Explain the various approaches to solving an LPP using the graphical method.

8. What is an iso-cost line and an iso-profit line?
9. Explain how to use the iso-profit line in a graphical maximization problem.
10. What are the various assumptions of a linear programming problem?
11. If a linear programming problem has a feasible region, then it has an infinite number of solutions. Explain.
12. Show that the set of feasible solutions to a given LPP forms a convex set.
13. Show that any BFS to an LPP is an extreme point of the convex set of feasible solutions of the problem.
14. If the feasible solution to a given LPP is a polytope, then show that at least one extreme point of the polytope will give the optimum solution of the given LPP.
15. A firm manufactures three products A, B, and C on which the profits earned per unit are \$10, \$15, and \$14, respectively. Each product is processed on two machines  $M_1$  and  $M_2$ . Product A requires one minute of processing time on  $M_1$  and three minutes on  $M_2$ , B requires two minutes on  $M_1$  and one minute on  $M_2$ , while C requires one minute on  $M_1$  and one minute on  $M_2$ . Machine  $M_1$  is available for not more than 15 hours, while  $M_2$  is available for 18 hours during any working day. Formulate this problem as an LP model so as to get maximum profit.
16. A shopkeeper plans to sell two types of products at costs of \$1150 and \$1600. The \$1150 product yields a profit of \$90 and the \$1600 product yields a profit of \$100. The shopkeeper estimates that the total monthly demand will not exceed 500 units. Find the number of units of each model that should be stocked in order to maximize profit. Also, it is given that the shopkeeper does not want to invest more than \$1,000,000 in inventory.
17. A company manufactures two types of models  $M_1$  and  $M_2$ . Each  $M_1$  model requires 4 hours of grinding and 2 hours of polishing, while each  $M_2$  model requires 2 hours of grinding and 5 hours of polishing. The company has 2 grinders and 3 polishers. Each grinder works for 40 hours a week and each polisher works for 60 hours a week. Profit on  $M_1$  is \$30 and that on a  $M_2$  model is \$40. Formulate this as an LPP and solve it graphically. A company manufactures two types of models  $M_1$  and  $M_2$ . Each  $M_1$  model requires 4 hours of grinding and 2 hours of polishing, while each  $M_2$  model requires 2 hours of grinding and 5 hours of polishing. The company has 2 grinders and 3 polishers. Each grinder works for

40 hours a week, and each polisher works for 60 hours a week. Profit on  $M_1$  is \$30, and that on an  $M_2$  model is \$40. Formulate this as an LPP and solve it graphically.

- 18.** The standard weight of a special purpose brick is 5 kg, and it contains two basic ingredients  $B_1$  and  $B_2$ .  $B_1$  costs \$5/ kg and  $B_2$  costs, \$8/ kg. Strength considerations dictate that the brick contains not more than 4 kg of  $B_1$  and a minimum of 2 kg of  $B_2$ . Since the demand for the product is likely to be related to the price of the brick, find graphically the minimum cost of the brick satisfying the previous conditions. Further, find the range of the ratio of costs of two materials so that the same solution remains optimal.
- 19.** A company has received a contract to supply gravel to three new construction projects located in towns A, B, and C. The construction engineers have estimated that the required amount of gravel which will be needed at these construction projects are:

| Project location | Weekly requirement (truck loads) |
|------------------|----------------------------------|
| A                | 72                               |
| B                | 102                              |
| C                | 41                               |

The company has 3 gravels pits located in towns X, Y, and Z. The gravel required by the construction projects can be supplied by three pits. The amount of gravel that can be supplied by each pit is as follows:

| Plant                          | X  | Y  | Z  |
|--------------------------------|----|----|----|
| Available amount (truck loads) | 76 | 82 | 77 |

The company has computed the delivery cost from each pit to each project site. These costs (in \$) are shown in the following table:

|     |   |    | Location |    |
|-----|---|----|----------|----|
|     |   | A  | B        | C  |
|     | X | 4  | 8        | 8  |
| Pit | Y | 16 | 24       | 16 |
|     | Z | 8  | 16       | 24 |

Schedule the shipment from each point to each project in such a manner that it minimizes the total transportation cost within the constraints imposed by pit capacities and project requirements. Formulate this transportation problem as a linear programming problem to minimize the total cost of transportation.

- 20.** A company has a scheduling problem. Operators are needed according to the schedule shown in the table. The operators report at the company at the beginning of each period and work for 8 consecutive hours. Let  $x_j$  equal the number of operators beginning work in time period  $j$ ,  $j = 1, 2, 3, \dots, 6$ . Formulate an LP model to hire the minimum number of operators the company needs.

| Time period        | Operators needed |
|--------------------|------------------|
| Midnight to 3 A.M. | 5                |
| 3 A.M. to 7 A.M.   | 10               |
| 7 A.M. to noon     | 88               |
| Noon to 3 P.M.     | 76               |
| 3 P.M. to 7 P.M.   | 50               |
| 7 P.M. to midnight | 15               |

- 21.** A factory manufactures three products which are processed through three different production stages. The time required for manufacturing one unit of each of the three products and the daily capacity of the stages are given in the following table:

| Stage           | Time required per unit |           |           | Stage Capacity |
|-----------------|------------------------|-----------|-----------|----------------|
|                 | Product 1              | Product 2 | Product 3 |                |
| 1               | 1                      | 1         | 1         | 430            |
| 2               | 3                      | ---       | 2         | 460            |
| 3               | 1                      | 4         | ---       | 420            |
| Profit per unit | 3                      | 2         | 5         |                |

Formulate the previous problem as a linear programming problem.

- 22.** A finished product must weigh exactly 150gms. The two raw materials used in manufacturing the product are A, with a cost of \$2 per unit, and

B, with a cost of \$8 per unit. At least 14 units of B and not more than 20 units of A must be used. Each unit of A and B weights 5 and 10 grams respectively. How much of each type of raw material should be used for each unit of the final product in order to minimize the cost?

23. A company produces two types of cowboy hats. Each hat of the first type requires twice as much labor time as does each hat of the second type. If all hats are of the second type only, the company can produce a total of 500 hats a day. The market limits daily sales of the first and second types to 150 and 200 hats. Assume that the profit per hat is \$8 for type 1 and \$5 for type 2. Determine the number of hats of each type to produce to maximize profit.
24. A firm has 240, 370, and 180 kg of wood, plastic, and steel, respectively. The firm produces two products A and B each unit respectively. Each unit of A requires 1, 3, and 2 kg of wood, plastic, and steel, respectively. The corresponding requirement for each unit of B is 3, 4, and 1 kg, respectively. If A sells for \$4 and B sells for \$6 per unit, then what product mix should the firm produce in order to have maximum gross income? Formulate this as an LPP.
25. Use the corner point approach to solve the following linear programming problems graphically:

(i)  $\text{Min} \quad z = 3x_1 + 5x_2$

Subject to

$$x_1 + x_2 \leq 3$$

$$0.5x_1 + 0.5x_2 \leq 1, x_1, x_2 \geq 0$$

(ii)  $\text{Max} \quad z = 5x_1 + 3.1x_2$

Subject to

$$3x_1 + 3.3x_2 \leq 30$$

$$x_2 \leq 6, x_1 \leq 7$$

$$x_1, x_2 \geq 0$$

(iii)  $\text{Min} \quad z = 5x_2$

Subject to

$$x_1 + 3.3x_2 \leq 10$$

$$x_1 + 10x_2 \geq 20, x_1, x_2 \geq 0$$

$$(iv) \quad \text{Max} \quad z = 1.5x_1 + 1.5x_2$$

Subject to

$$x_1 + 2x_2 \leq 10$$

$$3x_1 + 3x_2 \geq 12$$

$$x_2 \leq 3, x_1, x_2 \geq 0$$

$$(v) \quad \text{Min} \quad z = -x_1 - x_2$$

Subject to

$$-2x_1 + 2x_2 \leq 2$$

$$x_1 - 2x_2 \geq -4, x_1, x_2 \geq 0$$

$$(vi) \quad \text{Max} \quad z = 5x_1 + x_2$$

Subject to

$$3x_1 + 4x_2 \leq 20$$

$$3x_1 + 3x_2 \geq 21$$

$$x_2 \leq 3, x_1, x_2 \geq 0$$

**26.** Consider the following linear programming formulation:

$$\text{Min} \quad z = x_1 + 4x_2$$

Subject to

$$4x_1 + 2x_2 \geq 60$$

$$3x_1 + 10x_2 \geq 120$$

$$x_1 + 1.5x_2 \geq 30$$

$$\text{and} \quad x_1, x_2 \geq 0$$

Graphically illustrate the feasible region and apply the iso-cost line to find the optimal solution.

**27.** Discuss the following situations using the graphical method:

- (i) Unbounded solution
- (ii) Multiple solution
- (iii) Inconsistent constraint equations
- (iv) Redundancy